Theorem Inventory: Established Results of the Sedenion Program

Statements only — no proofs, no citations. Standing conventions: $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$ (Cayley–Dickson), τ the involution negating the odd half; $e_0,\ldots,e_7$ the octonion units with e_4 the mediator, $P = \{e_1,e_2,e_3\}$, $X = \{e_5,e_6,e_7\}$; L_x, R_x left/right multiplication; two-term zero divisors ("diagonals") $e_i \pm e_{8+u}$; $J_3(\mathbb{O})$ the exceptional Jordan algebra; all numerical claims verified in two independent implementations. Items marked ● are, to our knowledge, original; ○ marks results recovered independently that coincide with or extend known work.

Revision note. G3 and G4 (drafted and boarded during Program ω Stage C) restored — absent from the previous copy of this list, source undetermined (stale file or board regression). Two further Stage C/interpretation-session facts added (G5, G6). Two new sections added in full: H (the symplectic anatomy of the anchored dynamics, closing the branch-sector correspondence completely) and I (the wall's modular structure, including charge conjugation derived as a theorem). Lemmas are not included by standing decision; this remains a theorem-grade list. **Second pass**: A4's own restoration clause was checked directly rather than assumed correct — it claimed the kernel's exceptional 24-anchor set commutes with both τ and L_{e_4}; tested directly, τ-non-commutation is universal, all 84, no exceptions (only the L_{e_4} half has the 24-anchor exception G3 describes). Corrected. C4's cross-reference to H2 was a mismatch (the 8-family count is the census/Schur argument, unrelated to H2's associator reduction) — repointed to B1. G5's marker changed to ○/● per the A6/G1 convention, since the underlying finite-triviality result is textbook and only its application here is original.

A. The algebra $\mathbb{S}$ and its seam

A1 ● (Terminality.) Im $\mathbb{O}$ under the commutator is the unique non-Lie Malcev algebra arising in the Hurwitz chain; $\mathbb{S}$ is the last Cayley–Dickson double whose even part retains it. One further doubling leaves no Malcev-representable side.

A2 ● (Non-closure of the odd half.) For all $m, n \in \mathbb{O}e_8$: $mn \in \mathbb{O}$. The odd half has no self-product, hence admits no representation of itself; it is a faithful, irreducible, non-alternative Cayley bimodule over $\mathbb{O}$.

A3 ● (Seam localization.) Every two-term zero divisor of $\mathbb{S}$ has one index in each half; e_0 occurs in none; and for distinct nonzero j, k, the product $e_j e_k$ is never proportional to e_j or e_k.

A4 ● (Tunnel lemma.) For any two-term zero divisor A, both diagonal blocks of $T_A = L_A^T L_A$ equal $N(A)\cdot\mathrm{Id}$ exactly; the three-band spectrum $\{0^4, N(A)^8,$

$2N(A)^4$} resides entirely in the half-mixing cross-block, whose singular values are {N(A)×4, 0×4}. The kernel is exactly τ-balanced yet commutes with neither τ nor L_{e_4} — **generic, not universal**: the exceptional set is exactly the 24 extreme-class diagonals (G3), where the kernel is L_{e_4}-invariant; the τ-non-commutation is universal, all 84, no exceptions (τ maps ker L_{e_i+e_j} to ker L_{e_i−e_j}, the partner diagonal's own kernel, never its own — checked directly, deviation identical across all three classes, generic and both extreme).

A5 ● (Scalar-avoiding kernel.) For every zero divisor A on the full G_2-locus — not only the discrete diagonals — ker L_A is orthogonal to $\mathbb{R}e_0 \oplus \mathbb{R}(e_0e_8)$.

A6 ○/● (Strut characterization.) (i, u) is an assessor plane iff e_i e_u = ±e_S for a single unit e_S, and S is then the kite's strut; each of the seven kites omits exactly one Fano direction, the same in low and high halves, giving a bijection kites ↔ imaginary directions. (The seven box-kites of twelve diagonals, degree 4, recover known structure; the product-rule characterization and double-omission symmetry we state as found here.)

A7 ● (Directed strut fibration.) S(A) := dir Im($a \cdot \bar{b}$) for A = (a, b) is G_2-equivariant, equals −s·ε·e_S on every diagonal (ε evaluated on the grading-ordered pair), and satisfies S(b, a) = −S(a, b). Under color (the mediator's stabilizer), exactly the fibers over $\pm e_4$ are preserved; all others move.

A8 ● (The algebra's PT.) φ = diag(1,1,1,1,−1,−1,−1,−1) applied identically to both halves is an exact automorphism of $\mathbb{S}$ commuting with τ; it conjugates L_{e_4} to −L_{e_4}. Each kite's diagonals are uniformly φ-fixed or φ-exchanged, and the character equals the φ-parity of the strut: the P-strutted three are PT-even, the mediator- and X-strutted four are PT-odd.

A9 ● (Orbit and label theorem.) The four-group {id, τ, φ, τφ} preserves every index-plane; all orbits on the 84 diagonals are the diagonal-pairs. The label s·ε splits the 84 exactly 42/42, is flipped by τ, and equals minus the fiber sign of the directed strut.

B. Gauge structure from ($\mathbb{S}$, τ)

B1 ● (The gauge chain.) The derivation structure of $J_3(\mathbb{O})$ compatible with the split's data yields exactly su(3) ⊕ su(2) ⊕ u(1) with the Standard Model's charge assignments; in particular the lepton-to-quark doublet hypercharge ratio is exactly −3, derived from the SU(3) → SU(2)×U(1) branching with no phenomenological input.

B2 ● (Complex-structure census.) Exactly four gauge-compatible complex structures exist on the matter module: ±J_L ⊕ ±J_Q, forming an SM-shaped class and its mirror. Among the seven left-multiplications L_{e_i}, only L_{e_4} commutes with color; J_0 = −L_{e_4} (globally, in the pinned convention).

B3 ● (The gate theorem.) No operator on the module is simultaneously Ω-odd and T_3-commuting: any mass-type coupling that gaps the spectrum necessarily fails to commute with weak isospin. Fermion mass requires electroweak breaking, as a theorem of the algebra.

C. Chirality

C1 ● (Forced chirality.) The domain-wall zero-mode sector of the derived family (velocity splitter $\mathrm{Id} + \nu\Omega$, $\nu > 1$, mass K' from the (antisym, Ω-odd) gauge-covariant cell) carries content $\{1\_y\ell\} \oplus \{3\_yq\}$ that is necessarily chiral under unbroken $SU(3) \times U(1)\_Y$ — forced by Schur's lemma and $3 \ncong \bar{3}$, independent of all spectral detail.

C2 ● (Trade-off.) The alternative mass candidate $K = \Omega J_0$ ($K^2 = -\mathrm{Id}$, gauge-Cartan-commuting) is a removable pure twist: no gap, no wall. Chirality is purchased exactly at the price $[K', T_3] \neq 0$.

C3 ● (Wall anatomy.) The wall's eight real modes form four J_0-complex lines (one lepton, three quark) with internal weights $(\nu\pm1)/2\nu$; the compression of K obeys $P\_W\, K\, P\_W = (1/\nu)\cdot J_0|\_W$ exactly, with alignment vector $(+1, -1, -½, +½)$ across the four orientation classes.

C4 ● (Conjugation covariance.) Flow reversal acts on the orientation set as the global flip, forced by charge-lattice preservation; the quark block admits no color-commuting antilinear conjugation (complex type), the lepton block an 8-parameter family — the family's own size derived from the matter module's census (B1): the quark block is $\mathbb{C}^3\otimes\mathbb{C}^2$ under color, an intertwiner between 3 and $\bar{3}$, zero by Schur; the lepton doublet $\mathbb{C}^2$ has color acting trivially, so the family is every unconstrained antilinear map on $\mathbb{C}^2$, real dimension $2n^2=8$.

D. Protection of the orientation

D1 ● (The protection theorem.) On the closed operator algebra of legal perturbations, the orientation is unreachable by four jointly exhaustive mechanisms: (i) every symmetric-internal perturbation compresses to exactly zero on the wall (real bound profiles annihilate skew diagonals); (ii) every J_0-odd perturbation has alignment identically zero (compression preserves oddness, polar parts of odd operators are odd, traces of odd operators vanish); (iii) every even product of the derived-sign odd operators is a per-line scalar; (iv) every alignable operator's sign is convention or gauge — the weak Weyl element flips Ω and K exactly with J_0 invariant, and the T_3-conjugation orbit sweeps the K'-cell at uniform frequency 2λ, tracing $\cos(s)$ and reaching $-K'$ at $s = \pi/(2\lambda)$.

D2 ● (Closure.) The partition is multiplicatively closed: $\{K, K'\} = 0$ with $K\cdot K'$ landing in the K'-cell; $J_0\cdot K'$ is symmetric; $K\cdot J_0 = -\Omega$; commutators of cell elements lie in the knob span. Products cannot leave the partition.

D3 ● (Arrow-lock.) The framework's one derived sign — the thermal arrow, carried by A_sq — is J_0-odd and hence trace-locked away from the orientation at every perturbative order; the even power A_sq^2 = P_quark is a signless projector. Time's direction provably cannot select the mirror.

D4 ● (Verdict.) The orientation of chirality is a state-historical datum: selected by no derived structure, with the non-quasi-free stratum the only candidate author.

E. Seam-to-wall transport

E1 ● (One-channel transport.) A canonical, theorem-forced transport exists from the zero-divisor structure to the wall's orientations on exactly one channel: the mediator fiber's single bit onto ±J_Q, via fiber-sign → directed strut → complex structure. Every other channel is doubly excluded (gauge-incompatible by B2's uniqueness; remnant-swept where anything remains), and the lepton-relative sign lies outside every invariant space tested.

E2 ● (Remnant collapse.) The SO(3) remnant of color fixes the mediator fiber's directed strut exactly and sweeps every other fiber to its antipode; a remnant-invariant decoration of the seam carries exactly one bit — {±J_Q} — and no remnant-invariant lepton-relative sign exists.

E3 ● (Partner-pairing dichotomy.) The mediator kite is the unique kite whose assessor pairs are the module's own J_0-partner pairs (6/6 within-line); every other kite pairs across lines (0/6). **This same relationship is distinguished twice more, at two different levels of the construction**: it is what makes the mediator kite class-pure under the composition rule (G4), and what makes its diagonals symplectic hybrids rather than one of the two ordinary types (H1) — one quaternionic fact, visible at three levels.

F. States and quantization

F1 ● (Vacua addresses.) The CAR algebra over the wall with the mediator-partner pairing has Fock spectrum exactly binomial ($\{-4, -2^4, 0^6, +2^4, +4\}$); the four orientation classes sit at the four maximally polarized states (ℓ, q) = (±1, ±3), each the one-dimensional joint kernel of its own orientation's four lowering operators and of no other's.

F2 ● (Thermal blindness.) The wall Hamiltonian vanishes identically, so ρ_wall = Id/16 at every temperature: the unique thermal path is orientation-blind, and its zero-temperature limit is the mixed average, not a vacuum. The pure gauge-compatible ground states are exactly the four addressed vacua. **This same fact is the state whose Tomita–Takesaki structure is characterized completely in I1–I2**: the wall's thermal blindness is, in modular language, a state with no modular time at all.

F3 ● (Fermionization.) Under the derived contraction, the odd half's diagonal self-products vanish while cross-products survive: the sector acquires square-zero generator (Grassmann) structure. Fermionic statistics is derived from the flow, not imposed.

G. Settling

G1 ○/● (Depth conservation, scoped.) The depth coordinate χ, built from the symplectic spectrum of the covariance operator, is invariant under every symplectic (Bogoliubov) transformation; any depth-changing dynamics is necessarily non-symplectic. Independently, each quasi-free family member is a fixed point of the derived flow. (The invariance is classical; its role as the settling obstruction we state as found here.)

G2 ● (The carrier theorem.) For the mechanism class "singular scaling limit anchored at the zero-divisor locus": the exact Haar average over the locus gives $\langle T\_A \rangle = 2\cdot\mathrm{Id}$ and $\langle L\_A \rangle = 0$, with $\langle P\_kernel \rangle$ = the scalar/vector type split $\{0^2, (2/7)^{14}\}$ — direction-free members are orientation-contentless. Contentful members require an anchor; the anchor's fiber sign is carried faithfully (an s-flip leaves the diagonal blocks identical and negates exactly the cross-block) through the transport chain to $\pm J\_Q$. **Settling, in this class, is a carrier of the orientation and not an author of it**; nothing in the framework supplies the anchor.

G3 ● (The kernel-invariance criterion.) For any canonical diagonal anchor A, the tunnel kernel ker L_A is J_0-invariant iff L_c and L_a (the J_0-complex-linear and antilinear parts of L_A) annihilate it jointly and separately — automatic where $L\_c \equiv 0$, genuinely nontrivial at the mixed extreme class, failing on all 60 generic diagonals. Corollary, with Re(M)-annihilation of the kernel (universal across all 84): exactly the 24 extreme-class anchors carry a 4-dimensional frozen core — zero drift and zero diffusion — under the anchored Lindblad dynamics.

G4 ● (The composition rule.) Each basis unit carries an individual, cross-talk-free $(\|L\_c\|^2, \|L\_a\|^2)$ contribution under the $J_0 = -L\_\{e_4\}$ split: e_4 gives (16,0) (it *is* $-J_0$); e_{12} gives (0,16) (exact anticommutation); generic $\mathrm{Im}(\mathbb{O})$ gives (0,16) (via the established $C\ell(8)$ relation $\{L\_\{e\_i\}, L\_\{e\_j\}\} = -2\delta\_ij$, $i,j \le 8$); generic $\mathbb{O}e_8$ gives (8,8). Every diagonal anchor's class is the halved sum of its two ingredients — reproducing the 12/60/12 census identically, identifying the extreme pairs as exactly the e_4- and e_{12}-containing diagonals, and excluding the fourth combination (the $\{e_4, e_{12}\}$ pairing is never an assessor plane, since $e_4 \cdot e_4 = -e_0$). The three-class taxonomy is derived, not surveyed.

G5 ○/● (Sharp realization, resolved in the finite case.) Does the framework's own construction force a physically realized world to carry a definite orientation/anchor value? Resolved **no**, as a theorem, in any finite truncation (Shale–Stinespring triviality: every quasi-free state is quasi-equivalent to every other on a finite-mode CCR/CAR algebra, so orientation superpositions remain legal states, not artifacts of an incomplete construction). Open in the thermodynamic limit, where the deciding criterion is named exactly: orientation sectors become disjoint iff the mirror-flip acts with positive density across infinitely many modes. (Shale–Stinespring triviality is textbook; its application to orientation-sector realizability specifically is this program's own.)

G6 ● (Scalar-channel liveness.) The scalar directions (e_0, e_8) — which the kernel avoids identically and universally (A5) — are nonetheless dynamically live under the anchored dissipation: both carry nonzero drift and nonzero diffusion, confirmed on a representative of all three anchor classes, with no exception found. The dissipation does not conserve the measurement channel; record-stability, where the framework has it, is supplied by algebraic structure (the type-III folium), not by any dynamical shielding of the scalars themselves.

H. The symplectic anatomy of the anchored dynamics

H1 ● (The three-type classification, complete.) All 84 canonical diagonals carry exactly one of three symplectic anatomies under the anchored Lindblad dynamics, one invariant deciding which — the anchor's associator relationship with the mediator, (e_i, e_4, e_{j_low}). **48 generic diagonals pair fully across the seam**: the drift/diffusion-invariant decomposition (a middle band and the coupled kernel+top-band sector) is a genuine Lagrangian pair under the algebra's own symplectic form, with the flow's two observed branches given by a geometric mean $\nu(t)^2 = \mathrm{eig}(\sigma\_\mathrm{mid}(t)\cdot B\cdot\sigma\_\mathrm{coupled}(t)\cdot B^T)$ across the pairing, confirmed to four decimals against measured branches. **24 extreme-class diagonals decompose into three fully independent per-band phase spaces**, each individually Ω-nondegenerate and J_0-invariant, its own restricted symplectic spectrum reproducing the anchor's own measured branches exactly with no pairing needed. **The remaining 12 (the mediator kite) are a symplectic hybrid**: the middle band is self-contained (the per-band mechanism, locally), while the kernel and top band are Lagrangian duals of each other (the pairing mechanism, locally) — one anchor running both known mechanisms at once, partitioned by band, confirmed both statically (all four defining conditions exact) and dynamically (an integrated flow's composite prediction matching its own measured branches exactly). **This classification is strictly finer than the algebraic one**: all 60 non-extreme diagonals are the same class, (4,12), under the composition rule (G4) — the symplectic anatomy is invisible to it.

H2 ● (The associator computes what it classifies.) The cross-term driving the entire classification is T_A − T_i − T_j = −{L_i, L_j} (the mixed-half anticommutator), reducing completely to octonion-level operators: the relevant off-diagonal block is −[L_a, R_b]∘K (K the octonion conjugation), and [L_a, R_b] = −associator(a, ·, b) exactly — the same associator invariant that decides which of the three types an anchor belongs to is, up to sign and conjugation, the literal operator generating the mechanism that makes it so. The exact ratio between the two Lagrangian-pairing regimes ($\|[T\_A,J_0]\| = 8$ versus $8\sqrt{2}$) is derived by direct counting on this reduced structure: the driving commutator's anticommutator with L_4 is either 8 off-diagonal entries of magnitude 2 (non-partner, $\sqrt{(8\cdot 4)} = 4\sqrt{2}$) or 4 diagonal entries of magnitude 4 (partner, $\sqrt{(4\cdot 16)} = 8$), decided by whether the commutator's own invariant 4-dimensional subspace contains e_4 itself.

I. Modular structure of the wall

I1 ● (The tracial modular structure, and the shadow.) The wall's own thermal state (F2) is tracial, and Tomita–Takesaki theory for this state is exact and complete: the GNS space is the Hilbert–Schmidt matrices on $\mathbb{C}^{16}$, the modular operator $\Delta = 1$ (the state carries no modular time at all — a second, independent register for F2's own thermal blindness), and the modular conjugation is $J(X) = X^\dagger$. Extracted via the GNS one-particle sector (doubled, as the trace's own infinite-temperature structure requires) and factoring out the left–right exchange: the physical one-particle shadow is clean particle-hole conjugation on the wall's Majorana generators, $a\_k \leftrightarrow a\_k^\dagger$, with no residual string dependence.

I2 ● (Charge conjugation as a derived theorem.) The shadow of I1, restricted to the quark block and compared against the module's own color action: $c\,U\,c^{-1} = U^*$ for every $U \in SU(3)$ — the wall's modular conjugation implements standard charge conjugation on color, every color charge negated. This is forced, not assumed: C4's own theorem (no color-commuting antilinear conjugation exists on the quark block, since $3 \ncong \bar{3}$) requires exactly this failure of color-commutation from any antilinear map defined there, and the shadow supplies it as a genuine consequence of the state's own thermal structure rather than a postulated symmetry.

The one-sentence summary the list supports: from a single algebra and one involution, the program derives the gauge group with exact charges, the matter module with derived fermionic statistics, forced chirality, a complete protection theorem for the mirror choice, a one-bit bridge from the algebra's seam to the wall's orientations, the state-level vacuum structure, a complete symplectic classification of the one candidate carrier dynamics with its numerics derived to the last digit, charge conjugation obtained as a theorem of the wall's own thermal state rather than postulated — and a proof that the last remaining choice is an initial condition, carried but never made by anything the mathematics builds.